

MGMT 467 Executive Brief

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Looker Studio: <https://lookerstudio.google.com/s/iyh7BvyXLRc>

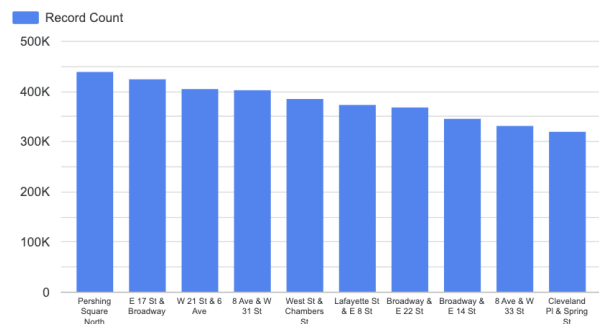
Context

For this project, we decided to analyze the Citi Bike collection of data, and with this data be able to find possible recommendations that we could give to Citi Bike to improve their engagement and bike availability. To be able to do all of this, we used Google Colab, BigQuery, Gemini, and LookerStudio. In the following report, we will be presenting and explaining our findings using the graphics that we made.

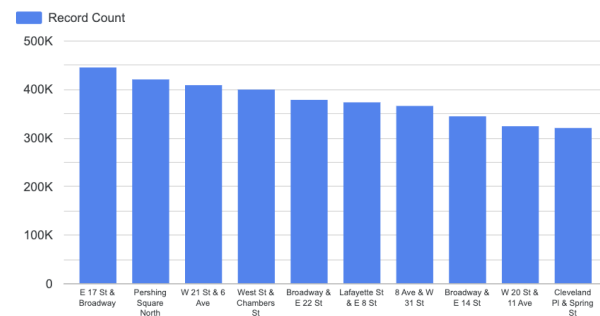
Step #1: Popular Stations

The first thing that we had to do, was find out which were the most widely used starting and ending stations, so that we had an idea of where there was going to be the most activity. So, what we did was create a bar chart that would be able to show us the total count of the top ten most used stations. First, we have the chart that shows us where most bikes are being taken out of, meaning that these are the stations where the greatest number of users are starting their ride. Second, we created the graph to show us the stations where the greatest number of bikes were being docked, ending stations. Having this information is crucial if we want to be able to find improvements to bike availability, if we want to ensure that customers have the highest probability of finding available bikes, we need to know where people are picking them up and leaving them.

Top 10 Starting Station



Top 10 Ending Station

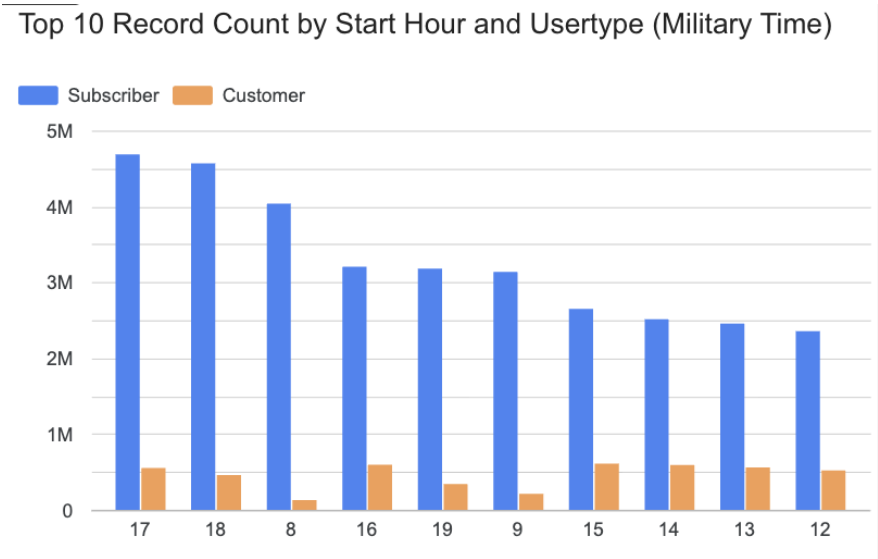
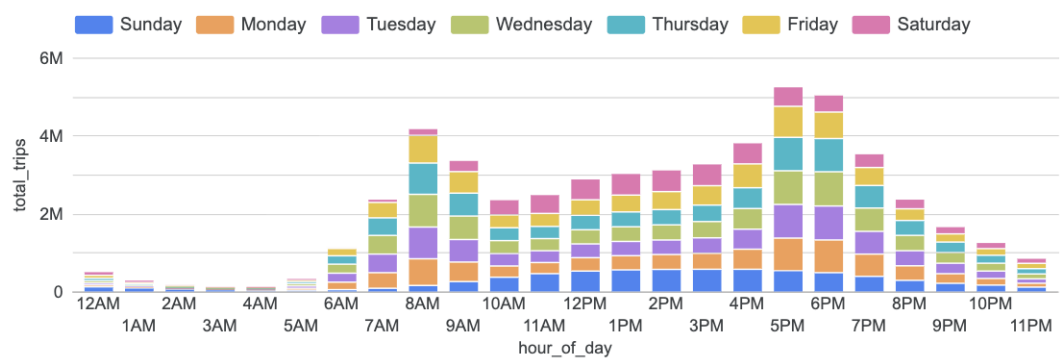


After looking at the graphs, we could see that the top 3 starting and ending stations were both the same, meaning that for our recommendations, we must keep these three at the front, and make sure that we take into account their importance.

Step #2: Peak Hours

Another big piece of information needed to make the necessary process recommendations was to look at the most popular hours that the bikes were used. If we want to make recommendations on availability, we need to know when people need to use the bikes, if not, we will not be able to improve the process. To find this information out, we created two different graphs.

Total Trips by Hour of Day and Day of Week

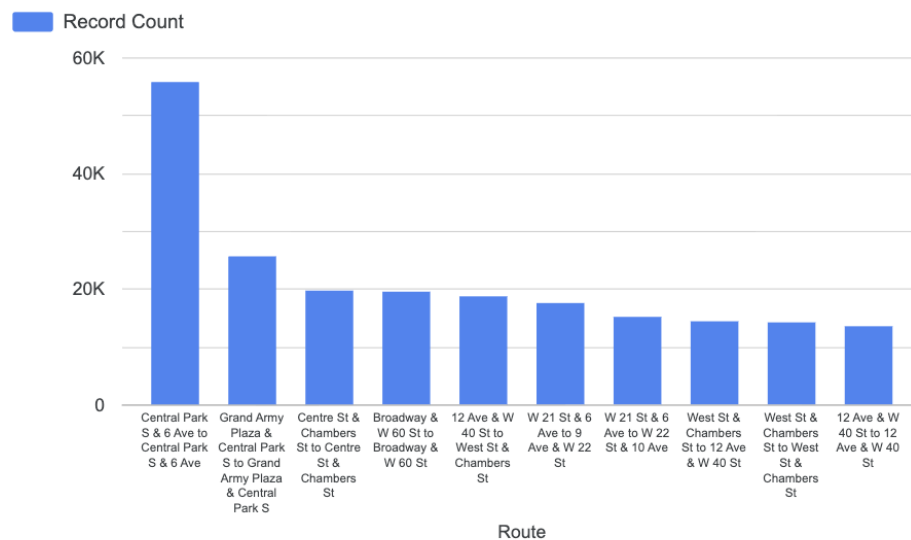


The first graph depicts the hours of the day, from midnight to 11 pm, where you can see the total number of rides starting at each hour, and it is divided into different days as well, so

we have an even clearer view. From what we can tell in the graphs, the most common hours in the morning are 8 and 9, and in the afternoon they are 5 and 6. We can deduce that these high numbers come from people either going to work or coming back home from work. To help us make the right recommendations, this is a very important statistic. We can also see that for workdays, the number of trips during working hours are very consistent, but when you look at the weekends, those same consistent hours change. You will see that on weekends, there is a gradual increase from 8 am to 2 pm, and then a decrease from 2 pm to 10 pm, another important factor when making our recommendations.

Step #3: Favorite Routes

Top 10 Record Count by Route (Starting Station to Ending Station)



Another analysis that we decided to do was to see the top routes that people were taking. We believed that by doing this, and combining it with the information that we have about the top starting and ending locations, we could make better improvements. If we know the routes that people are taking, we can use it to try and make sure that in these stations there are always bikes ready. Furthermore, knowing where people are passing by, we can get an idea of, if we were to add more biking docks, where we would need to add them.

Step #4: Recommendations

Based on our analysis, our group has identified several key recommendations for the NYC Department of Transportation to improve Citi Bike usage and operational efficiency. First, we recommend ensuring that the most frequently used start and end stations are fully stocked with bikes prior to the morning and evening rush hours. By redistributing additional

bikes to these high-demand areas, Citi Bike can help ensure consistent availability for riders and reduce shortages during peak commute times. Second, we recommend expanding the dock capacity at the three most popular stations to improve overall traffic flow. Since the top three stations for both starting and ending trips are the same, it's clear that these locations experience consistently high volumes of activity. Increasing their capacity, combined with targeted bike redistribution, would help reduce congestion and ensure smoother operations during peak hours. Third, we suggest scheduling bike maintenance and system downtime during low-demand periods, such as mid-day on weekdays or early mornings on weekends. Performing repairs and rebalancing during these off-peak hours will help maximize bike availability when demand is highest and minimize disruptions to riders.